

## Homogeneous Catalysis | Hot Paper |

 A Convenient Palladium-Catalyzed Carbonylative Suzuki Coupling of Aryl Halides with Formic Acid as the Carbon Monoxide SourceXinxin Qi,<sup>[a]</sup> Li-Bing Jiang,<sup>[a]</sup> Hao-Peng Li,<sup>[a]</sup> and Xiao-Feng Wu<sup>\*[a, b]</sup>

Dedicated to Professor Pierre H. Dixneuf for his outstanding scientific contributions

**Abstract:** A practical palladium-catalyzed carbonylative Suzuki coupling of aryl halides under carbon monoxide gas-free conditions has been developed. Here, formic acid was utilized as the carbon monoxide source for the first time

with acetic anhydride as the additive. A variety of diarylketones were produced in moderate to excellent yields from the corresponding aryl halides and arylboronic acids.

## Introduction

Diarylketone is one of the most important types of building blocks that is frequently present in natural products, pharmaceutical compounds, and other biologically active molecules.<sup>[1]</sup> Many synthetic methods for diarylketones synthesis have been reported. One typical strategy is Friedel–Crafts acylation of arenes.<sup>[2]</sup> However, the requirement for greater than stoichiometric amounts of Lewis acid (AlCl<sub>3</sub>) and low regioselectivity are the main drawbacks. Hence, alternative synthetic approaches have been developed, including Fries rearrangement,<sup>[3]</sup> acylation of benzoic acid derivatives with organometallic reagents,<sup>[4]</sup> among others.<sup>[5]</sup>

Among all known methodologies for diarylketone synthesis, transition-metal-catalyzed three-component coupling of aryl halides, organometallic species, and carbon monoxide (CO) offers an ideal and promising option.<sup>[6,7]</sup> This strategy is regarded as an efficient and straightforward tool for diarylketone construction because two carbon–carbon bonds are formed in a single step and a variety of aryl metal reagents, including aluminum,<sup>[8]</sup> magnesium,<sup>[9]</sup> silicon,<sup>[10]</sup> tin,<sup>[11]</sup> and boron<sup>[12]</sup> can be applied. However, the usage of gaseous CO limits the applications of carbonylations in laboratories because CO gas is toxic, flammable, and odorless. As part of our continuing interest in developing new palladium-catalyzed carbonylation reactions, herein we wish to report a convenient and general approach for the synthesis of ketones under CO-free conditions. By palladium-catalyzed carbonylative Suzuki coupling of aryl halides and boronic acids, with formic acid<sup>[13]</sup> as the CO source and

acetic anhydride as the additive, the desired diarylketones were isolated in moderate to excellent yields. Notably, this procedure represents the first example of carbonylative Suzuki coupling with formic acid as the liquid CO source.

## Results and Discussion

First, the carbonylative cross-coupling reaction was conducted by using iodobenzene and phenylboronic as substrates with palladium acetate as the catalyst, 1,2-bis(diphenylphosphino)ethane (DPPE) as the ligand, formic acid as the CO source, acetic anhydride as the additive, and K<sub>2</sub>CO<sub>3</sub> as the base in toluene at 100 °C. To our delight, 26% yield of benzophenone was obtained (Table 1, entry 1). Next, various ligands were studied (Table 1, entries 1–10). For the bidentate ligands, DPPPE gave the highest yield (Table 1, entry 4); DPPF provided a similar yield to that of DPPE (Table 1, entry 5), and the others resulted in lower yields. Gratifyingly, when we chose the monodentate ligand PPh<sub>3</sub>, the final product was obtained in 82% yield, whereas PCy<sub>3</sub> resulted in 8% yield (Table 1, entries 9 and 10). Then, different bases were tested (Table 1, entries 10–16), and K<sub>2</sub>CO<sub>3</sub> still gave the best result. In the investigation into the effects of solvents, toluene was the optimal solvent (Table 1, entries 17–19). Additionally, the amounts of base and phenylboronic acid, reaction temperature, and catalyst loading were all varied and no better results were observed.

Based on these promising results, we next studied the influence of different formic acid activators. As shown in Table 2, compared with acetic acid, none of the tested anhydrides could provide the desired ketone under our best reaction conditions.

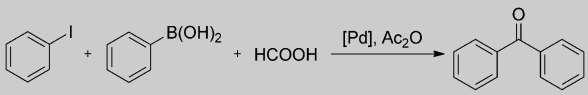
The optimized reaction conditions were then applied to test the applicability of this carbonylative Suzuki coupling reaction for a range of substrates. A variety of aryl iodides were tested in the first stage, as shown in Table 3, for substrates with electron-donating groups, such as methyl, trimethyl, and methoxy groups, to afford the diarylketone products in moderate to good yields (Table 3, entries 2–6). Interestingly, the amino

[a] Dr. X. Qi, L.-B. Jiang, H.-P. Li, Prof. Dr. X.-F. Wu  
Department of Chemistry, Zhejiang Sci-Tech University  
Xiasha Campus, Hangzhou 310018 (P.R. China)  
E-mail: xiao-feng.wu@catalysis.de

[b] Prof. Dr. X.-F. Wu  
Leibniz-Institut für Katalyse e.V. an der Universität Rostock  
Albert-Einstein-Strasse 29a, 18059 Rostock (Germany)

Supporting information for this article is available on the WWW under <http://dx.doi.org/10.1002/chem.201502943>.

**Table 1.** Screening of reaction conditions.<sup>[a]</sup>

|  |                       |                                 |                    |                          |
|-----------------------------------------------------------------------------------|-----------------------|---------------------------------|--------------------|--------------------------|
| Entry                                                                             | Ligand <sup>[c]</sup> | Base                            | Solvent            | Yield [%] <sup>[b]</sup> |
| 1 <sup>[c]</sup>                                                                  | DPPE                  | K <sub>2</sub> CO <sub>3</sub>  | toluene            | 26                       |
| 2 <sup>[c]</sup>                                                                  | DPPP                  | K <sub>2</sub> CO <sub>3</sub>  | toluene            | 5                        |
| 3 <sup>[c]</sup>                                                                  | DPPB                  | K <sub>2</sub> CO <sub>3</sub>  | toluene            | 13                       |
| 4 <sup>[c]</sup>                                                                  | DPPPE                 | K <sub>2</sub> CO <sub>3</sub>  | toluene            | 33                       |
| 5 <sup>[c]</sup>                                                                  | DPPF                  | K <sub>2</sub> CO <sub>3</sub>  | toluene            | 27                       |
| 6 <sup>[c]</sup>                                                                  | BINAP                 | K <sub>2</sub> CO <sub>3</sub>  | toluene            | 15                       |
| 7 <sup>[c]</sup>                                                                  | xantphos              | K <sub>2</sub> CO <sub>3</sub>  | toluene            | 0                        |
| 8 <sup>[c]</sup>                                                                  | X-Phos                | K <sub>2</sub> CO <sub>3</sub>  | toluene            | 2                        |
| 9                                                                                 | PCy <sub>3</sub>      | K <sub>2</sub> CO <sub>3</sub>  | toluene            | 8                        |
| 10                                                                                | PPh <sub>3</sub>      | K <sub>2</sub> CO <sub>3</sub>  | toluene            | 82                       |
| 11                                                                                | PPh <sub>3</sub>      | Na <sub>2</sub> CO <sub>3</sub> | toluene            | 6                        |
| 12                                                                                | PPh <sub>3</sub>      | K <sub>3</sub> PO <sub>4</sub>  | toluene            | 30                       |
| 13                                                                                | PPh <sub>3</sub>      | <i>t</i> BuOK                   | toluene            | 34                       |
| 14                                                                                | PPh <sub>3</sub>      | KOAc                            | toluene            | 6                        |
| 15                                                                                | PPh <sub>3</sub>      | Et <sub>3</sub> N               | toluene            | 5                        |
| 16                                                                                | PPh <sub>3</sub>      | DBU                             | toluene            | 16                       |
| 17                                                                                | PPh <sub>3</sub>      | K <sub>2</sub> CO <sub>3</sub>  | anisole            | 28                       |
| 18                                                                                | PPh <sub>3</sub>      | K <sub>2</sub> CO <sub>3</sub>  | DMF                | 13                       |
| 19                                                                                | PPh <sub>3</sub>      | K <sub>2</sub> CO <sub>3</sub>  | 1,4-dioxane        | 2                        |
| 20                                                                                | PPh <sub>3</sub>      | K <sub>2</sub> CO <sub>3</sub>  | CH <sub>3</sub> CN | 5                        |
| 21                                                                                | PPh <sub>3</sub>      | K <sub>2</sub> CO <sub>3</sub>  | H <sub>2</sub> O   | 0                        |

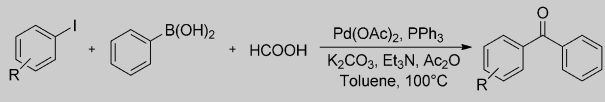
[a] Reaction conditions: iodobenzene (1.0 mmol), phenylboronic acid (1.2 mmol), Pd(OAc)<sub>2</sub> (3 mol%), ligand (6 mol%), base (3 mmol), Et<sub>3</sub>N (4 mmol), HCOOH (3.0 mmol), acetic anhydride (3.0 mmol), solvent (6 mL), 100 °C, 12 h. [c] DPPP: 1,3-bis(diphenylphosphino)propane; DPPB: 1,4-bis(diphenylphosphino)butane; DPPPE: 1,5-bis(diphenylphosphino)pentane; DPPF: 1,1'-bis(diphenylphosphino)ferrocene; BINAP: 2,2'-bis(diphenylphosphino)-1,1'-binaphthyl; xantphos: 4,5-bis(diphenylphosphino)-9,9-dimethylxanthene; X-Phos: 2-dicyclohexylphosphino-2',4',6'-triisopropylbiphenyl; Cy: cyclohexyl; DBU: 1,5-diazabicyclo[5.4.0]undec-5-ene. [b] Yield determined by GC, with dodecane as an internal standard. [c] Ligand (4 mol%).

**Table 2.** Influences of different CO sources.<sup>[a]</sup>

| Entry | Anhydrides                | Yield [%] <sup>[b]</sup> |
|-------|---------------------------|--------------------------|
| 1     | propanoic anhydride       | 0                        |
| 2     | butyric anhydride         | 0                        |
| 3     | succinic anhydride        | 0                        |
| 4     | maleic anhydride          | 0                        |
| 5     | BOC anhydride             | 0                        |
| 6     | phthalic anhydride        | 0                        |
| 7     | benzoic anhydride         | 3                        |
| 8     | trifluoroacetic anhydride | 0                        |

[a] Reaction conditions: iodobenzene (1.0 mmol), phenylboronic acid (1.2 mmol), Pd(OAc)<sub>2</sub> (3 mol%), PPh<sub>3</sub> (6 mol%), K<sub>2</sub>CO<sub>3</sub> (3 mmol), Et<sub>3</sub>N (4 mmol), HCOOH (3.0 mmol), anhydrides (3.0 mmol), toluene (6 mL), 100 °C, 10–12 h. [b] Yield determined by GC, with dodecane as an internal standard. BOC anhydride: di-*tert*-butyl dicarbonate.

**Table 3.** Carbonylative Suzuki coupling reaction of aryl iodides with phenylboronic acid.<sup>[a]</sup>

|  |             |              |                          |
|------------------------------------------------------------------------------------|-------------|--------------|--------------------------|
| Entry                                                                              | Aryl iodide | Diarylketone | Yield [%] <sup>[b]</sup> |
| 1                                                                                  |             |              | 78                       |
| 2                                                                                  |             |              | 65                       |
| 3                                                                                  |             |              | 68                       |
| 4                                                                                  |             |              | 54                       |
| 5                                                                                  |             |              | 73                       |
| 6                                                                                  |             |              | 78                       |
| 7                                                                                  |             |              | 25                       |
| 8                                                                                  |             |              | 40                       |
| 9                                                                                  |             |              | 74                       |
| 10                                                                                 |             |              | 31                       |
| 11                                                                                 |             |              | 55                       |
| 12                                                                                 |             |              | 46                       |
| 13                                                                                 |             |              | 41                       |
| 14                                                                                 |             |              | 82                       |
| 15                                                                                 |             |              | 88                       |
| 16                                                                                 |             |              | 75                       |
| 17                                                                                 |             |              | 77                       |

group can be tolerated under our conditions and gave the desired ketone in 25% yield (Table 3, entry 7). Both fluoro and chloro substituents are unreacted and gave the desired substituted ketones in moderate to good yields (Table 3, entries 8–

| Table 3. (Continued) |             |              |                          |
|----------------------|-------------|--------------|--------------------------|
| Entry                | Aryl iodide | Diarylketone | Yield [%] <sup>[b]</sup> |
| 18                   |             |              | 69                       |
| 19                   |             |              | 57                       |
| 20                   |             |              | 26                       |
| 21                   |             |              | 80                       |
| 22                   |             |              | 43                       |
| 23                   |             |              | 68                       |

[a] Reaction conditions: aryl iodides (1.0 mmol), phenylboronic acid (1.2 mmol), Pd(OAc)<sub>2</sub> (3 mol%), PPh<sub>3</sub> (6 mol%), K<sub>2</sub>CO<sub>3</sub> (3 mmol), Et<sub>3</sub>N (4 mmol), HCOOH (3.0 mmol), acetic anhydride (3.0 mmol), toluene (6 mL), 100 °C, 10–12 h. [b] Isolated yield.

11). Here, *ortho*- and *meta*-chloro groups afforded the diarylketones in lower yields than that in the *para*-substituted case (Table 3, entry 9 versus 10 and 11). The nitro group is usually not tolerated in CO gas-based carbonylation because it will be reduced and further transformed. Surprisingly, 41% of 4-nitro ketone was isolated under our conditions without further optimization (Table 3, entry 13). Other electron-withdrawing groups, such as ester, trifluoro, and nitrile groups, gave the desired products in excellent yields as well (Table 3, entries 14–16). Notably, biphenyl and naphthalene substrates can react smoothly to provide the corresponding products in high yields (Table 3, entries 17–19). Moreover, heteroaryl iodides were also investigated and moderate to good yields of the corresponding products could be achieved (Table 3, entries 22 and 23).

With these results in hand, we turned our attention to the formation of diarylketones with aryl bromides (Table 4). Under the same reaction conditions, both electron-rich- and -poor-substituted aryl bromides can provide the corresponding diarylketones with phenylboronic acid in moderate yield.

Further investigation of the substrate scope was carried out by using a variety of arylboronic acids under standard reaction conditions (Table 5). When substrates with electron-donating moieties, such as methyl, methoxy, trifluoromethoxy, and 3,5-dimethyl groups, good to excellent yields of the diarylketones were obtained (Table 5, entries 1, 4, and 5). Those substrates with *meta*- and *ortho*-substituted aryl groups gave products in lower yields than those with *para* substitution (Table 5, entry 1 vs. 2 and 3; 5 vs. 6 and 7). We also checked halide groups: chloro substitution afforded the desired product in 77% yield (Table 5, entry 10), whereas the fluoro group resulted in 36%

| Table 4. Carbonylative Suzuki coupling reaction of aryl bromides with phenylboronic acid. <sup>[a]</sup> |              |              |                          |
|----------------------------------------------------------------------------------------------------------|--------------|--------------|--------------------------|
|                                                                                                          |              |              |                          |
| Entry                                                                                                    | Aryl bromide | Diarylketone | Yield [%] <sup>[b]</sup> |
| 1                                                                                                        |              |              | 55                       |
| 2                                                                                                        |              |              | 55                       |
| 3                                                                                                        |              |              | 45                       |
| 4                                                                                                        |              |              | 64                       |
| 5                                                                                                        |              |              | 15                       |
| 6                                                                                                        |              |              | 56                       |
| 7                                                                                                        |              |              | 46                       |

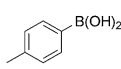
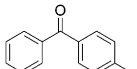
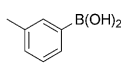
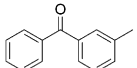
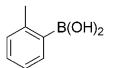
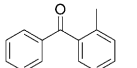
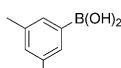
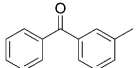
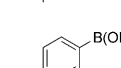
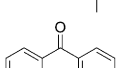
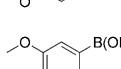
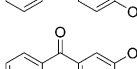
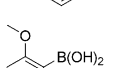
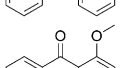
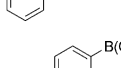
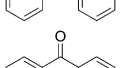
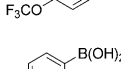
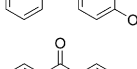
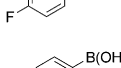
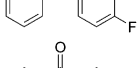
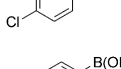
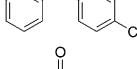
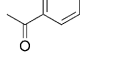
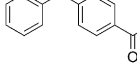
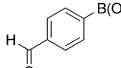
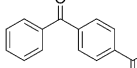
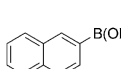
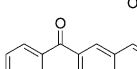
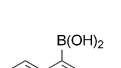
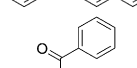
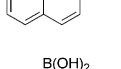
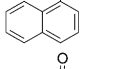
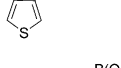
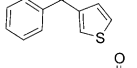
[a] Reaction conditions: aryl bromides (1.0 mmol), phenylboronic acid (1.2 mmol), Pd(OAc)<sub>2</sub> (3 mol%), PPh<sub>3</sub> (6 mol%), K<sub>2</sub>CO<sub>3</sub> (3 mmol), Et<sub>3</sub>N (4 mmol), HCOOH (3.0 mmol), acetic anhydride (3.0 mmol), toluene (6 mL), 100 °C, 10–12 h. [b] Yield of product isolated.

yield (Table 5, entry 9). It was noted that the electronic influence of the substituents had a profound impact on the yield. Electron-deficient groups provided the diarylketones in relatively poor yields compared with electron-donating groups; this is due to the decomposition of activated arylboronic acids (Table 5, entries 11 and 12). In addition, the reaction can also smoothly proceed in the presence of naphthalene groups to give the products in good yields (Table 5, entries 13 and 14). The reaction of methyl 4-iodobenzoate with (4-methoxyphenyl)boronic acid and *p*-tolylboronic acid smoothly proceeded to give the corresponding diarylketones in 58 and 61% yield, respectively (Table 5, entries 16 and 17). 4-Iodobenzonitrile and *p*-tolylboronic acid can also be applied and provide the desired product in 65% yield (Table 5, entry 18).

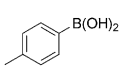
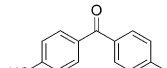
In addition, the carbonylative Suzuki coupling reaction was also conducted with 1-bromo-4-iodobenzene and phenylboronic acid to give 30% yield of 1,4-phenylenebis(phenylmethane) without further optimization (Scheme 1).

Concerning the reaction mechanism, a plausible pathway is proposed and shown in Scheme 2. The catalytic cycle starts with the oxidative addition of aryl halides to {Pd<sup>0</sup>L<sub>n</sub>} to give an arylpalladium complex and then forms an acylpalladium intermediate after the coordination and insertion of CO. The final diarylketone will be eliminated after reductive elimination and return palladium(0) for next catalyst cycle. Herein, CO is in situ

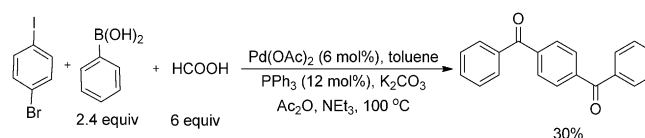
**Table 5.** Carbonylative Suzuki coupling reaction of iodobenzene with arylboronic acid.<sup>[a]</sup>

| Entry | Aryl boronic acid                                                                   | Diarylketone                                                                        | Yield [%] <sup>[b]</sup> |
|-------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------|
| 1     |    |    | 84                       |
| 2     |    |    | 59                       |
| 3     |    |    | 62                       |
| 4     |    |    | 85                       |
| 5     |    |    | 83                       |
| 6     |    |    | 56                       |
| 7     |    |    | 30                       |
| 8     |   |   | 56                       |
| 9     |  |  | 36                       |
| 10    |  |  | 77                       |
| 11    |  |  | 23                       |
| 12    |  |  | 21                       |
| 13    |  |  | 62                       |
| 14    |  |  | 66                       |
| 15    |  |  | 35                       |
| 16    |  |  | 58                       |
| 17    |  |  | 61                       |

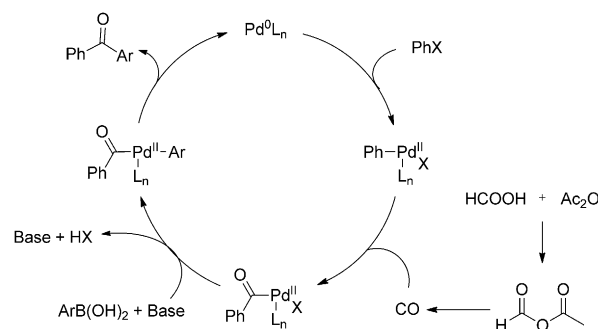
**Table 5.** (Continued)

| Entry | Aryl boronic acid                                                                  | Diarylketone                                                                        | Yield [%] <sup>[b]</sup> |
|-------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------|
| 18    |  |  | 65                       |

[a] Reaction conditions: iodobenzene (1.0 mmol), arylboronic acid (1.2 mmol), Pd(OAc)<sub>2</sub> (3 mol%), PPh<sub>3</sub> (6 mol%), K<sub>2</sub>CO<sub>3</sub> (3 mmol), Et<sub>3</sub>N (4 mmol), HCOOH (3.0 mmol), acetic anhydride (3.0 mmol), toluene (6 mL), 100 °C, 10–12 h. [b] Yield of product yield.



**Scheme 1.** Palladium-catalyzed carbonylative Suzuki reaction of 1-bromo-4-iodobenzene.



**Scheme 2.** Proposed mechanism for the coupling reaction.

produced from acetic formic anhydride, which is obtained in situ by the reaction of formic acid and acetic anhydride.

## Conclusion

We have developed a practical and convenient synthetic method for diarylketones. Through palladium-catalyzed carbonylative Suzuki coupling reactions of aryl halides with arylboronic acids, with formic acid as the CO source, moderate to good yields of the desired diarylketones could be obtained. This strategy can avoid the usage of gaseous CO. Remarkably, this methodology represents the first report of using formic acid as a liquid CO source for the carbonylative Suzuki reaction.

## Experimental Section

### Typical reaction procedure for carbonylative Suzuki coupling reactions

Pd(OAc)<sub>2</sub> (3 mol%) and PPh<sub>3</sub> (6 mol%) were transferred into an oven-dried tube filled with nitrogen. Toluene (6.0 mL), arylhalides

(1.0 mmol), arylboronic acid (1.2 mmol),  $K_2CO_3$  (3 mmol), and  $Et_3N$  (4 mmol) were added into the reaction tube. Then a mixture of formic acid (3.0 mmol) and acetic anhydride (3.0 mmol), which was stirred for 1.5 h at 30 °C, was added dropwise to the reaction tube. The mixture was stirred for 10–12 h at 100 °C. After the reaction was complete, the reaction mixture was filtered and concentrated and purified by column chromatography on silica gel (petroleum ether/ethyl acetate 40:1).

## Characterization

**Benzophenone:**  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  = 7.80 (d,  $J$  = 7.7 Hz, 4H), 7.58 (t,  $J$  = 7.4 Hz, 2H), 7.47 ppm (t,  $J$  = 7.6 Hz, 4H);  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  = 196.8, 137.6, 132.4, 130.0, 128.3 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 182.0 (6)  $[M]^+$ , 182.0 (77), 105.0 (100), 77.0 (50), 51.0 (14).

**Phenyl(*o*-tolyl)methanone:**  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  = 7.83 (d,  $J$  = 7.7 Hz, 2H), 7.60 (t,  $J$  = 7.4 Hz, 1H), 7.50–7.25 (m, 6H), 2.36 ppm (s, 3H);  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  = 198.5, 138.5, 137.6, 136.6, 133.0, 130.9, 130.1, 130.0, 128.39, 128.35, 125.1, 19.9 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 196.0 (11)  $[M]^+$ , 195.1 (100), 178.1(11), 185.1(10), 119.0 (22), 105.0 (12), 91.0 (25), 77.0 (23), 65.0 (11), 51.0 (10).

**Phenyl(*m*-tolyl)methanone:**  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  = 7.79 (d,  $J$  = 7.7 Hz, 2H), 7.63 (s, 1H), 7.56 (dd,  $J$  = 9.2, 2.8 Hz, 2H), 7.46 (t,  $J$  = 7.6 Hz, 2H), 7.36 (dt,  $J$  = 14.9, 7.5 Hz, 2H), 2.40 ppm (s, 3H);  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  = 196.8, 138.0, 137.6, 137.5, 133.1, 132.2, 130.3, 129.9, 128.1, 128.0, 127.3, 21.2 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 196.0 (10)  $[M]^+$ , 196.1 (100), 181.0 (25), 165.0 (11), 152.1 (11), 119.0 (90), 105.0 (60), 91.0 (10), 77.0 (40), 65.0 (15), 51.0 (10).

**4-Methylbenzophenone:**  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  = 7.66 (d,  $J$  = 7.7 Hz, 2H), 7.60 (d,  $J$  = 8.0 Hz, 2H), 7.45 (dd,  $J$  = 13.6, 6.7 Hz, 1H), 7.34 (t,  $J$  = 7.6 Hz, 2H), 7.15 (d,  $J$  = 7.9 Hz, 2H), 2.30 ppm (s, 3H);  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  = 196.2, 143.0, 137.7, 134.7, 132.2, 132.0, 130.1, 129.8, 129.7, 128.8, 128.0, 21.4 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 196.0 (12)  $[M]^+$ , 196.1 (92), 181.0 (17), 165.0 (10), 152.0 (10), 119.0 (100), 105.0 (30), 91.1 (32), 77.0 (30), 65.0 (14), 51.0 (10).

**[4-(*tert*-Butyl)phenyl](phenyl)methanone:**  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  = 7.80 (d,  $J$  = 7.8 Hz, 2H), 7.76 (d,  $J$  = 8.2 Hz, 2H), 7.56 (t,  $J$  = 7.3 Hz, 1H), 7.49 (d,  $J$  = 8.0 Hz, 3H), 7.45 (d,  $J$  = 7.7 Hz, 2H), 1.36 ppm (s, 9H);  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  = 196.4, 156.1, 137.9, 134.7, 132.1, 130.1, 129.9, 128.1, 125.2, 35.0, 31.1 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 238.0 (13)  $[M]^+$ , 238.1 (40), 223.1 (100), 195.1 (10), 161.1 (20), 105.0 (30), 77.0 (24).

**(4-Methoxyphenyl)(phenyl)methanone:**  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  = 7.80 (d,  $J$  = 8.7 Hz, 2H), 7.73 (d,  $J$  = 7.5 Hz, 2H), 7.53 (t,  $J$  = 7.3 Hz, 1H), 7.43 (t,  $J$  = 7.6 Hz, 2H), 6.93 (d,  $J$  = 8.7 Hz, 2H), 3.83 ppm (s, 3H);  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  = 195.3, 163.1, 138.1, 132.4, 131.7, 129.9, 129.5, 128.0, 113.4, 55.3 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 212.0 (10)  $[M]^+$ , 212.1 (67), 135.0 (100), 105.0 (13), 92.0 (13), 77.0 (25).

**(4-Aminophenyl)(phenyl)methanone:**  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  = 7.70 (d,  $J$  = 5.5 Hz, 4H), 7.53 (t,  $J$  = 7.2 Hz, 1H), 7.44 (t,  $J$  = 7.4 Hz, 2H), 6.84 (d,  $J$  = 8.0 Hz, 2H), 4.52 ppm (s, 2H);  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  = 195.7, 150.7, 138.6, 135.6, 133.0, 132.6, 131.5, 129.5, 128.1, 127.9, 127.4, 113.9 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 197.0 (9)

$[M]^+$ , 197.0 (80), 120.0 (100), 105.0 (10), 92.0 (21), 77.0 (20), 65.0 (20), 51.0 (10).

**(4-Fluorophenyl)(phenyl)methanone:**  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  = 7.83 (dt,  $J$  = 11.8, 5.9 Hz, 2H), 7.77 (t,  $J$  = 7.1 Hz, 2H), 7.68–7.56 (m, 3H), 7.48 (t,  $J$  = 7.7 Hz, 3H), 7.41–7.32 (m, 1H), 7.20–7.12 ppm (m, 3H);  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  = 195.4, 165.4 (d,  $J$  = 253.6 Hz), 137.5, 132.7, 132.6, 132.5, 129.9, 128.4, 115.5 ppm (d,  $J$  = 21.8 Hz); GC-MS (EI, 70 eV):  $m/z$  (%): 200.0 (10)  $[M]^+$ , 200.0 (100), 123.0 (85), 105.0 (54), 95.0 (35), 77.0 (31), 51.0 (10).

**(4-Chlorophenyl)(phenyl)methanone:**  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  = 7.82–7.72 (m, 4H), 7.59 (t,  $J$  = 7.2 Hz, 1H), 7.47 ppm (dd,  $J$  = 15.3, 7.6 Hz, 4H);  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  = 195.4, 138.8, 137.2, 135.8, 132.6, 131.4, 129.9, 128.6, 128.4 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 217.0 (8)  $[M]^+$ , 216.0 (100), 181.0 (18), 152.0 (12), 139.0 (59), 111.0 (24), 105.0 (100), 77.0 (41), 51.0 (12).

**(2-Chlorophenyl)(phenyl)methanone:**  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  = 7.84–7.77 (m, 2H), 7.66 (dd,  $J$  = 11.9, 7.6 Hz, 1H), 7.60–7.52 (m, 1H), 7.46 (dt,  $J$  = 12.1, 5.9 Hz, 4H), 7.35 ppm (d,  $J$  = 12.5 Hz, 1H);  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ): 195.3, 138.6, 137.6, 136.5, 133.7, 132.5, 131.2, 130.1, 129.1, 128.6, 128.3, 126.7 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 217.0 (10)  $[M]^+$ , 216.0 (86), 181.0 (10), 152.0 (10), 139.0 (57), 111.0 (20), 105.0 (100), 77.0 (40), 51.1 (10).

**(3-Chlorophenyl)(phenyl)methanone:**  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  = 7.78 (d,  $J$  = 7.9 Hz, 3H), 7.68–7.53 (m, 3H), 7.49 (t,  $J$  = 7.6 Hz, 2H), 7.41 ppm (t,  $J$  = 7.8 Hz, 1H);  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  = 195.2, 139.2, 136.9, 134.5, 132.8, 132.3, 130.0, 129.9, 129.6, 128.4, 128.1 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 217.0 (12)  $[M]^+$ , 216.0 (65), 181.0 (12), 152.0 (10), 139.0 (30), 111.0 (24), 105.0 (100), 77.0 (35), 51.0 (12).

**1-(4-Benzoylphenyl)ethan-1-one:**  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  = 8.05 (d,  $J$  = 8.0 Hz, 2H), 7.84 (t,  $J$  = 9.2 Hz, 2H), 7.80 (d,  $J$  = 7.5 Hz, 2H), 7.61 (t,  $J$  = 7.3 Hz, 1H), 7.49 (t,  $J$  = 7.5 Hz, 2H), 2.66 ppm (s, 3H);  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  = 197.4, 195.8, 141.2, 139.5, 136.8, 132.9, 130.0, 129.9, 128.4, 128.1, 26.8 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 224.0 (14)  $[M]^+$ , 224.0 (58), 209.0 (100), 181.0 (10), 147.0 (25), 105.0 (42), 77.0 (25), 51.0 (10).

**(4-Nitrophenyl)(phenyl)methanone:**  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  = 8.32 (t,  $J$  = 8.9 Hz, 2H), 7.92 (d,  $J$  = 8.3 Hz, 2H), 7.79 (d,  $J$  = 7.7 Hz, 2H), 7.64 (t,  $J$  = 7.4 Hz, 1H), 7.50 ppm (dd,  $J$  = 16.5, 8.9 Hz, 2H);  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  = 194.8, 149.8, 142.8, 136.2, 133.4, 130.7, 130.1, 128.6, 123.6 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 227.0 (10)  $[M]^+$ , 227.0 (56), 197.0 (11), 181.0 (10), 150.0 (200), 120.0 (18), 105.0 (100), 92.0 (10), 77.0(44), 51.0 (11).

**Methyl 4-benzoylbenzoate:**  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  = 8.12 (d,  $J$  = 8.1 Hz, 2H), 7.79 (dd,  $J$  = 14.4, 7.9 Hz, 4H), 7.58 (t,  $J$  = 7.4 Hz, 1H), 7.47 (t,  $J$  = 7.6 Hz, 2H), 3.93 ppm (s, 3H);  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  = 195.9, 166.2, 141.2, 136.9, 133.1, 132.9, 130.0, 129.7, 129.4, 128.4, 52.4 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 240.0 (13)  $[M]^+$ , 240.0 (96), 209.0 (32), 181.0 (24), 163.0 (80), 152.0 (13), 135.0 (13), 105.0 (100), 77.0 (40), 51.0 (11).

**Phenyl[4-(trifluoromethyl)phenyl]methanone:**  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  = 7.88 (d,  $J$  = 8.0 Hz, 2H), 7.78 (t,  $J$  = 8.4 Hz, 2H), 7.74 (d,  $J$  = 8.0 Hz, 2H), 7.62 (t,  $J$  = 7.4 Hz, 1H), 7.50 ppm (t,  $J$  = 7.5 Hz, 2H);  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  = 195.5, 140.8, 136.7, 133.7 (q,  $J$  =



33.5 Hz), 133.1, 130.2, 130.1, 128.5, 125.4 (q,  $J=3.5$  Hz), 123.7 ppm (q,  $J=275.3$  Hz); GC-MS (EI, 70 eV):  $m/z$  (%): 250.0 (13)  $[M]^+$ , 250.0 (80), 231.0 (13), 181.0 (10), 173.0 (50), 145.0 (50), 102.0 (100), 95.0 (10), 77.0 (45), 51.0 (10).

**4-Benzoylbenzonitrile:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta=7.86$  (d,  $J=8.1$  Hz, 2H), 7.77 (dd,  $J=8.5$ , 2.1 Hz, 4H), 7.63 (t,  $J=7.3$  Hz, 1H), 7.50 ppm (t,  $J=7.6$  Hz, 2H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta=195.0$ , 141.2, 136.3, 133.3, 132.1, 130.2, 130.0, 128.6, 118.0, 115.6 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 207.0 (13)  $[M]^+$ , 207.0 (88), 130.0 (33), 105.0 (100), 77.0 (44), 51.0 (18).

**(1,1'-Biphenyl)-4-yl(phenyl)methanone:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta=7.93$ –7.87 (m, 2H), 7.84 (d,  $J=7.8$  Hz, 2H), 7.73–7.68 (m, 2H), 7.68–7.63 (m, 2H), 7.60 (t,  $J=7.4$  Hz, 1H), 7.54–7.44 (m, 4H), 7.41 ppm (t,  $J=7.3$  Hz, 1H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta=196.3$ , 145.2, 139.9, 137.7, 136.2, 132.3, 130.7, 130.0, 128.9, 128.3, 128.2, 127.3, 126.9 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 258.0 (15)  $[M]^+$ , 258.1 (56), 181.0 (100), 152.0 (39), 127.0 (11), 115.0 (17), 105.0 (22), 77.0 (22), 51.0 (10).

**Naphthalen-1-yl(phenyl)methanone:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta=8.10$  (d,  $J=8.2$  Hz, 1H), 8.00 (d,  $J=8.1$  Hz, 1H), 7.92 (d,  $J=7.6$  Hz, 1H), 7.87 (d,  $J=7.9$  Hz, 2H), 7.64–7.56 (m, 2H), 7.55–7.48 (m, 3H), 7.45 ppm (t,  $J=7.7$  Hz, 2H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta=195.5$ , 138.9, 137.6, 137.3, 135.9, 132.7, 132.4, 131.5, 130.1, 130.0, 128.7, 128.4, 128.3 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 232.0 (11)  $[M]^+$ , 232.0 (100), 215.0 (13), 202.0 (13), 155.0 (69), 127.0 (50), 105.0 (21), 77.0 (23), 51.0 (10).

**Naphthalen-2-yl(phenyl)methanone:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta=8.27$  (s, 1H), 8.03–7.84 (m, 6H), 7.60 (dt,  $J=13.7$ , 6.7 Hz, 2H), 7.57–7.46 ppm (m, 3H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta=196.7$ , 137.8, 135.2, 134.7, 132.3, 132.2, 131.8, 130.0, 129.3, 128.3, 128.2, 127.7, 126.7, 125.7 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 232.0 (10)  $[M]^+$ , 232.0 (100), 202.0 (10), 155.0 (954), 127.0 (59), 105.0 (27), 77.0 (27), 51.0 (10).

**Phenyl(thiophen-3-yl)methanone:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta=7.93$  (d,  $J=1.6$  Hz, 1H), 7.84 (d,  $J=7.6$  Hz, 2H), 7.58 (dd,  $J=9.8$ , 6.1 Hz, 2H), 7.48 (t,  $J=7.6$  Hz, 2H), 7.40–7.35 ppm (m, 1H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta=190.0$ , 141.3, 138.6, 133.9, 132.3, 129.4, 128.6, 128.4, 126.2 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 188.0 (9)  $[M]^+$ , 188.0 (100), 160.0 (13), 111.0 (90), 105.0 (38), 82.9 (13), 77.0 (33), 51.1 (13).

**Phenyl(pyridin-3-yl)methanone:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta=8.97$  (s, 1H), 8.79 (d,  $J=4.5$  Hz, 1H), 8.13 (d,  $J=7.8$  Hz, 1H), 7.77 (d,  $J=7.7$  Hz, 2H), 7.60 (t,  $J=7.3$  Hz, 1H), 7.47 ppm (dd,  $J=13.9$ , 6.7 Hz, 3H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta=194.4$ , 152.1, 150.2, 137.7, 136.5, 133.3, 133.2, 129.9, 128.6, 123.5 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 183.0 (12)  $[M]^+$ , 183.0 (100), 154.0 (10), 105.0 (60), 77.0 (35), 51.0 (15).

**Phenyl(pyridin-2-yl)methanone:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta=13.88$ –13.66 (m, 1H), 8.71 (d,  $J=4.7$  Hz, 1H), 8.03 (t,  $J=8.6$  Hz, 3H), 7.88 (t,  $J=7.7$  Hz, 1H), 7.57 (t,  $J=7.4$  Hz, 1H), 7.46 ppm (dd,  $J=9.9$ , 5.5 Hz, 3H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta=193.7$ , 154.8, 148.4, 137.1, 136.1, 132.9, 130.9, 128.1, 126.1, 124.6 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 183.0 (10)  $[M]^+$ , 182.0 (68), 155.0 (100), 105.0 (56), 77.0 (56), 51.0 (22).

**Phenyl(quinolin-6-yl)methanone:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta=9.08$  (d,  $J=2.5$  Hz, 1H), 8.36 (d,  $J=8.8$  Hz, 1H), 8.31–8.22 (m, 2H), 8.15 (d,  $J=8.8$  Hz, 1H), 7.82 (d,  $J=7.5$  Hz, 2H), 7.61 (t,  $J=7.4$  Hz, 1H), 7.49 ppm (dd,  $J=14.6$ , 8.6 Hz, 3H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta=195.7$ , 151.6, 148.4, 138.4, 137.1, 135.6, 132.7, 131.2, 130.0, 129.9, 128.9, 128.4, 127.6, 127.4, 121.9 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 233.0 (10)  $[M]^+$ , 233.0 (100), 186.0 (80), 128.0 (32), 105.0 (40), 77.0 (28), 51.0 (10).

**(3,5-Dimethylphenyl)(phenyl)methanone:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta=7.80$  (d,  $J=7.5$  Hz, 2H), 7.58 (t,  $J=7.4$  Hz, 1H), 7.47 (t,  $J=7.5$  Hz, 2H), 7.40 (s, 2H), 7.22 (s, 1H), 2.37 ppm (s, 6H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta=197.1$ , 137.87, 137.85, 137.6, 134.0, 132.2, 130.0, 128.2, 127.8, 21.2 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 210.0 (11)  $[M]^+$ , 210.0 (77), 195.0 (27), 133.0 (100), 105.0 (56), 77.0 (34), 51.0 (10).

**(3-Methoxyphenyl)(phenyl)methanone:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta=7.79$  (d,  $J=7.9$  Hz, 2H), 7.55 (t,  $J=7.4$  Hz, 1H), 7.45 (t,  $J=7.5$  Hz, 2H), 7.38–7.28 (m, 3H), 7.12 (t,  $J=9.5$  Hz, 1H), 3.82 ppm (s, 3H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta=196.3$ , 159.4, 138.7, 137.4, 132.3, 129.9, 129.1, 128.1, 122.7, 118.7, 114.2, 55.3 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 212.0 (13)  $[M]^+$ , 212.0 (100), 181.0 (10), 135.0 (73), 105.0 (60), 92.0 (10), 77.0 (45).

**(2-Methoxyphenyl)(phenyl)methanone:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta=7.80$  (d,  $J=7.9$  Hz, 2H), 7.54 (t,  $J=7.2$  Hz, 1H), 7.43 (dt,  $J=15.4$ , 7.7 Hz, 3H), 7.35 (d,  $J=7.4$  Hz, 1H), 7.06–6.95 (m, 2H), 3.70 ppm (s, 3H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta=196.4$ , 157.3, 137.7, 132.9, 131.8, 129.7, 129.5, 128.7, 128.1, 120.4, 111.4, 55.5 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 212.0 (15)  $[M]^+$ , 212.0 (66), 195.0 (34), 181.0 (10), 167.1 (13), 152.0 (10), 135.0 (100), 121.0 (26), 105.0 (29), 92.0 (17), 77.0 (43), 51.0 (10).

**Phenyl[4-(trifluoromethoxy)phenyl]methanone:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta=7.85$  (d,  $J=8.4$  Hz, 2H), 7.77 (d,  $J=7.7$  Hz, 2H), 7.58 (t,  $J=7.3$  Hz, 1H), 7.47 (t,  $J=7.6$  Hz, 2H), 7.30 ppm (d,  $J=8.4$  Hz, 2H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta=195.0$ , 152.1, 137.1, 135.8, 132.6, 131.9, 129.9, 128.4, 120.1, 119.3 ppm (q,  $J=257.3$  Hz); GC-MS (EI, 70 eV):  $m/z$  (%): 266.0 (15)  $[M]^+$ , 266.0 (84), 189.0 (100), 181.0 (17), 161.0 (15), 105.0 (58), 95.0 (21), 77.0 (25), 51.0 (10).

**4-Benzoylbenzaldehyde:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta=10.12$  (s, 1H), 7.99 (d,  $J=8.0$  Hz, 2H), 7.92 (d,  $J=8.1$  Hz, 2H), 7.81 (t,  $J=10.3$  Hz, 2H), 7.62 (t,  $J=7.4$  Hz, 1H), 7.49 ppm (dd,  $J=19.6$ , 12.0 Hz, 2H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta=195.8$ , 191.6, 142.6, 138.5, 136.7, 133.1, 130.3, 130.1, 129.5, 128.5 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 210.0 (12)  $[M]^+$ , 210.0 (77), 181.0 (22), 133.0 (40), 105.0 (100), 77.0 (40), 51.0 (12).

**Methyl 4-(4-methoxybenzoyl)benzoate:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta=8.09$  (d,  $J=8.2$  Hz, 2H), 7.75 (dd,  $J=12.1$ , 8.6 Hz, 4H), 6.92 (d,  $J=8.7$  Hz, 2H), 3.91 (s, 3H), 3.84 ppm (s, 3H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta=194.6$ , 166.2, 163.5, 142.0, 132.6, 132.5, 132.1, 129.4, 129.3, 113.6, 113.3, 55.4, 52.3 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 270.0 (11)  $[M]^+$ , 270.1 (55), 239.0 (11), 211.0 (10), 163.0 (10), 135.0 (100), 77.0 (10).

**Methyl 4-(4-methylbenzoyl)benzoate:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta=8.12$  (d,  $J=8.1$  Hz, 2H), 7.80 (d,  $J=8.1$  Hz, 2H), 7.70 (d,  $J=7.9$  Hz, 2H), 7.29 (t,  $J=7.9$  Hz, 2H), 3.94 (s, 3H), 2.43 ppm (s, 3H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta=195.4$ , 166.0, 143.6, 141.5, 134.0,

132.8, 130.1, 129.4, 129.2, 128.9, 52.2, 21.5 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 254.0 (15)  $[M]^+$ , 254.0 (67), 223.0 (22), 195.0 (20), 163.0 (25), 119.0 (100), 97.0 (25), 65.0 (10).

**4-(4-Methylbenzoyl)benzonitrile:**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta$  = 7.74 (d,  $J$  = 8.4 Hz, 2H), 7.67 (d,  $J$  = 8.4 Hz, 2H), 7.59 (d,  $J$  = 8.1 Hz, 2H), 7.20 (d,  $J$  = 8.0 Hz, 2H), 2.35 ppm (s, 3H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta$  = 194.5, 144.2, 141.5, 133.5, 131.9, 130.1, 129.9, 129.2, 117.9, 115.2 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 221.0 (12)  $[M]^+$ , 221.1 (56), 206.1 (10), 130.0 (12), 119.0 (100), 102.0 (17), 91.0 (33), 65.0 (12).

**1,4-Phenylenebis(phenylmethanone):**  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ ):  $\delta$  = 7.88 (s, 4H), 7.83 (d,  $J$  = 7.5 Hz, 4H), 7.62 (t,  $J$  = 7.4 Hz, 2H), 7.50 ppm (t,  $J$  = 7.5 Hz, 4H);  $^{13}\text{C}$  NMR (101 MHz,  $\text{CDCl}_3$ ):  $\delta$  = 196.0, 140.6, 136.9, 132.9, 130.1, 129.7, 128.5 ppm; GC-MS (EI, 70 eV):  $m/z$  (%): 286.0 (10)  $[M]^+$ , 286.1 (100), 209.0 (90), 181.0 (19), 152.0 (22), 105.0 (95), 77.0 (44), 51.0 (12).

## Acknowledgements

We are grateful for financial support from the NSFC (21472174), the Education Department of Zhejiang Province (Y201432060), and Zhejiang Sci-Tech University (1206838-Y; 14062015-Y). X.-F.W. appreciates the general support from Matthias Beller in LIKAT.

**Keywords:** carbonylation • cross-coupling • heterogeneous catalysis • palladium • synthesis design

- [1] a) Y. Jiang, P. Tu, *Chem. Pharm. Bull.* **2005**, *53*, 1164–1166; b) L.-H. D. Nguyen, G. Venkatraman, K.-Y. Sim, L. J. Harrison, *Phytochemistry* **2005**, *66*, 1718–1723; c) J. W. Lampe, C. K. Biggers, J. M. Defauw, R. J. Fogle-song, S. E. Hall, J. M. Heerding, S. P. Hollinshead, H. Hu, P. F. Hughes, G. E. Jagdmann, Jr., M. G. Johnson, Y.-S. Lai, C. T. Lowden, M. P. Lynch, J. S. Mendoza, M. M. Murphy, J. W. Wilson, L. M. Ballas, K. Carter, J. W. Darges, J. E. Davis, F. R. Hubbard, M. L. Stamper, *J. Med. Chem.* **2002**, *45*, 2624–2643; d) S. Rancon, A. Chaboud, N. Darbour, G. Comte, C. Bayet, P.-N. Simon, J. Raymond, A. Di Pietro, P. Cabalion, D. Barron, *Phytochem-istry* **2001**, *57*, 553–557; e) H. Ito, E. Nishitani, T. Konoshima, M. Takasaki, M. Kozuka, T. Yoshida, *Phytochemistry* **2000**, *54*, 695–700; f) J.-C. Li, T. Nohara, *Chem. Pharm. Bull.* **2000**, *48*, 1354–1355; g) N. De Kimpe, M. Keppens, G. Froncg, *Chem. Commun.* **1996**, 635–636.
- [2] a) C. E. Song, W. H. Shim, E. J. Roh, J. H. Choi, *Chem. Commun.* **2000**, 1695–1696; b) A. Fürstner, D. Voigtländer, W. Schrader, D. Giebel, M. T. Reetz, *Org. Lett.* **2001**, *3*, 417–420; c) S. J. Nara, J. R. Harjani, M. M. Sal-unkhe, *J. Org. Chem.* **2001**, *66*, 8616–8620; d) J. Ross, J. L. Xiao, *Green Chem.* **2002**, *4*, 129–133; e) S. Gmouh, H. Yang, M. Vaultier, *Org. Lett.* **2003**, *5*, 2219–2222; f) E. Fillion, D. Fishlock, A. Wilsily, J. M. Goll, *J. Org. Chem.* **2005**, *70*, 1316–1327; g) D. O. Jang, K. S. Moon, D. H. Cho, J. G. Kim, *Tetrahedron Lett.* **2006**, *47*, 6063–6066; h) G. Sartori, R. Maggi, *Chem. Rev.* **2006**, *106*, 1077–1104; i) A. Dzudza, T. J. Marks, *J. Org. Chem.* **2008**, *73*, 4004–4016.
- [3] a) D. Bellus, P. Hrdlovic, *Chem. Rev.* **1967**, *67*, 599–609; b) M. P. Sibi, V. Snieckus, *J. Org. Chem.* **1983**, *48*, 1935–1937.
- [4] a) J. W. Labadie, J. K. Stille, *J. Am. Chem. Soc.* **1983**, *105*, 6129–6137; b) L. J. Goossen, K. Ghosh, *Angew. Chem. Int. Ed.* **2001**, *40*, 3458–3460; *Angew. Chem.* **2001**, *113*, 3566–3568; c) Y. C. Huang, K. K. Majumdar, C.-H. Cheng, *J. Org. Chem.* **2002**, *67*, 1682–1684; d) A. Zapf, *Angew. Chem. Int. Ed.* **2003**, *42*, 5394–5399; *Angew. Chem.* **2003**, *115*, 5552–5557; e) D. Wang, Z. Zhang, *Org. Lett.* **2003**, *5*, 4645–4648; f) H. Tatamidani, K. Yokota, F. Kakiuchi, N. Chatani, *J. Org. Chem.* **2004**, *69*, 5615–5621; g) Y. Zhang, T. Rovis, *J. Am. Chem. Soc.* **2004**, *126*, 15964–15965; h) G. F. Silbestri, R. Bogel-Masson, M. T. Lockhart, A. B. Chopra, *J. Organomet. Chem.* **2006**, *691*, 1520–1524.
- [5] X. F. Wu, *Chem. Eur. J.* **2015**, *21*, 12252–12265.
- [6] For recent reviews on carbonylation, see: a) R. Grigg, S. P. Mutton, *Tetrahedron* **2010**, *66*, 5515–5548; b) X.-F. Wu, H. Neumann, *ChemCatChem* **2012**, *4*, 447–458; c) X.-F. Wu, H. Neumann, M. Beller, *ChemSusChem* **2013**, *6*, 229–241; d) X.-F. Wu, H. X. Fang, L. Wu, R. Jackstell, H. Neu-mann, M. Beller, *Acc. Chem. Res.* **2014**, *47*, 1041–1053; e) L. Wu, X. Fang, Q. Liu, R. Jackstell, M. Beller, X.-F. Wu, *ACS Catal.* **2014**, *4*, 2977–2989; f) Q. Liu, H. Zhang, A. Lei, *Angew. Chem. Int. Ed.* **2011**, *50*, 10788–10799; *Angew. Chem.* **2011**, *123*, 10978–10989.
- [7] For selected recent examples, see: a) Y. Zhao, L. Jin, P. Li, A. Lei, *J. Am. Chem. Soc.* **2008**, *130*, 9429–9433; b) H. Zhang, R. Shi, P. Gan, C. Liu, A. Ding, Q. Wang, A. Lei, *Angew. Chem. Int. Ed.* **2012**, *51*, 5204–5207; *Angew. Chem.* **2012**, *124*, 5294–5297; c) W. Li, C. Liu, H. Zhang, K. Ye, G. Zhang, W. Zhang, Z. Duan, S. You, A. Lei, *Angew. Chem. Int. Ed.* **2014**, *53*, 2443–2446; *Angew. Chem.* **2014**, *126*, 2475–2478; d) L. Wang, Y. Wang, C. Liu, A. Lei, *Angew. Chem. Int. Ed.* **2014**, *53*, 5657–5661; *Angew. Chem.* **2014**, *126*, 5763–5767; e) W. Li, Z. Duan, X. Zhang, H. Zhang, M. Wang, R. Jiang, H. Zeng, C. Liu, A. Lei, *Angew. Chem. Int. Ed.* **2015**, *54*, 1893–1896; *Angew. Chem.* **2015**, *127*, 1913–1916; f) K. T. Neumann, S. R. Laursen, A. T. Lindhardt, B. Bang-Andersen, T. Skrydstrup, *Org. Lett.* **2014**, *16*, 2216–2219; g) C. Brancour, T. Fukuyama, Y. Mukai, T. Skrydstrup, I. Ryu, *Org. Lett.* **2013**, *15*, 2794–2797; h) P. Xie, Y. Xie, B. Qian, H. Zhou, C. Xia, H. Huang, *J. Am. Chem. Soc.* **2012**, *134*, 9902–9905.
- [8] N. A. Bumagin, A. B. Ponomaryov, I. P. Beletskaya, *Tetrahedron Lett.* **1985**, *26*, 4819–4822.
- [9] Y. Yamamoto, T. Kohara, A. Yamamoto, *Chem. Lett.* **1976**, 1217–1220.
- [10] Y. Hatanaka, S. Fukushima, T. Hiyama, *Tetrahedron* **1992**, *48*, 2113–2119.
- [11] a) A. M. Echavarren, J. K. Stille, *J. Am. Chem. Soc.* **1988**, *110*, 1557–1565; b) S. K. Kang, T. Yamaguchi, T. H. Kim, P. S. Ho, *J. Org. Chem.* **1996**, *61*, 9082–9083; c) E. Morera, G. Ortar, *Bioorg. Med. Chem. Lett.* **2000**, *10*, 1815–1818; d) S. Ceccarelli, U. Piarulli, C. Gennari, *J. Org. Chem.* **2000**, *65*, 6254–6256.
- [12] a) T. Ohe, K. Ohe, S. Uemura, N. Sugita, *J. Organomet. Chem.* **1988**, *344*, C5–C7; b) T. Ishiyama, H. Kizaki, T. Hayashi, A. Suzuki, N. Miyauro, *J. Org. Chem.* **1998**, *63*, 4726–4731; c) M. B. Andrus, Y. Ma, Y. Zang, C. Song, *Tetrahedron Lett.* **2002**, *43*, 9137–9140; d) T. Ishiyama, H. Kizaki, N. Miyauro, A. Suzuki, *Tetrahedron Lett.* **1993**, *34*, 7595–7598; e) H. Neu-mann, A. Brennfürer, M. Beller, *Chem. Eur. J.* **2008**, *14*, 3645–3652.
- [13] a) S. Cacchi, G. Fabrizi, A. Goggiamani, *Org. Lett.* **2003**, *5*, 4269–4272; b) S. Cacchi, G. Fabrizi, A. Goggiamani, *J. Comb. Chem.* **2004**, *6*, 692–694; c) S. Korsager, R. H. Taaning, T. Skrydstrup, *J. Am. Chem. Soc.* **2013**, *135*, 2891–2894; d) J. Hou, J. H. Xie, Q.-L. Zhou, *Angew. Chem. Int. Ed.* **2015**, *54*, 6302–6305; *Angew. Chem.* **2015**, *127*, 6400–6403; e) X. Qi, L. B. Jiang, C. L. Li, R. Li, X. F. Wu, *Chem. Asian J.* **2015**, *10*, 1870–1873.

Received: July 27, 2015

Revised: September 16, 2015

Published online on October 21, 2015